

Recap lecture 5



- Different notations of transition diagrams, languages of strings of **even length**, **Odd length**, **starting with b**, **ending in a** (with different FAs), **beginning with a**, **not beginning with b**, **beginning with and ending in same letters**

TASK



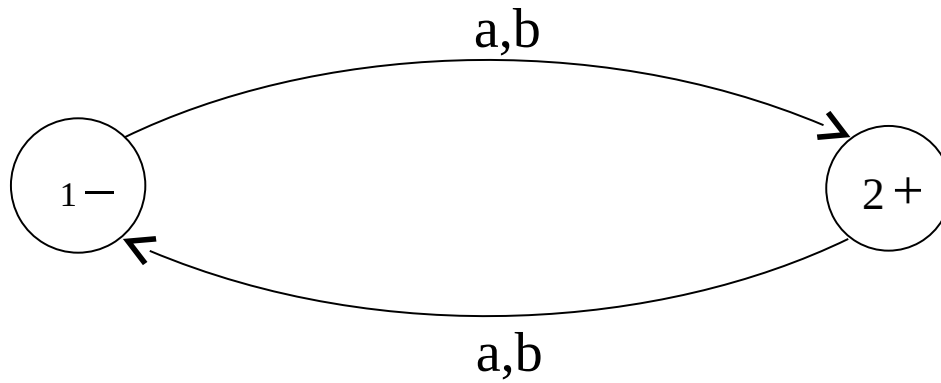
Build an FA for the language L of strings, defined over $\Sigma = \{a, b\}$, **of odd length.**

Solution: The language L may be expressed by RE

$(a+b)((a+b)(a+b))^*$ or
 $((a+b)(a+b))^*(a+b)$

This language may be accepted by the following FA

Solution continued ...



Task

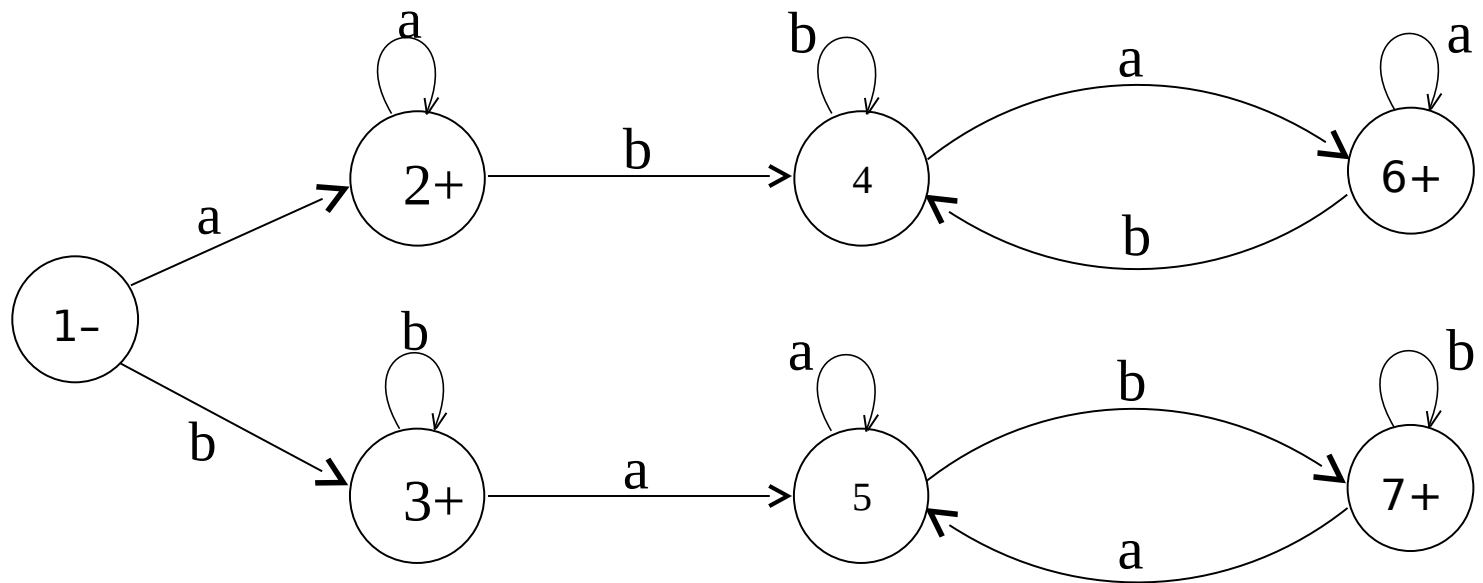
- Build an FA accepting the Language L of Strings, defined over $\Sigma = \{a, b\}$, **beginning with and ending in same letters.**

Solution: The language L may be expressed by the following regular expression

$$(a+b)+a(a+b)^*a + b(a+b)^*b$$

This language L may be accepted by the following FA

Solution continued ...



Example

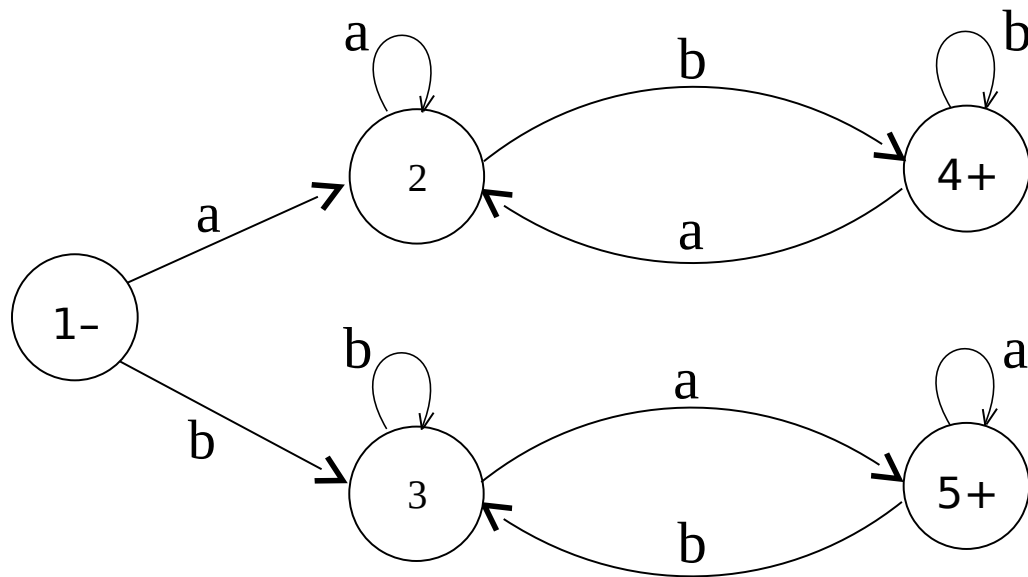
Consider the Language L of Strings , defined over $\Sigma = \{a, b\}$, **beginning with and ending in different letters.**

The language L may be expressed by the following regular expression

$$a(a + b)^*b + b(a + b)^*a$$

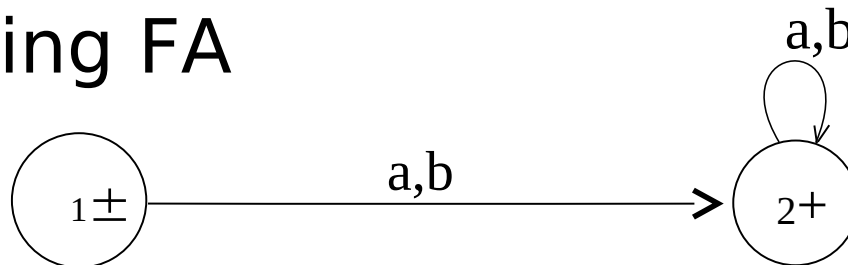
This language may be accepted by the following FA

Example Continued ...



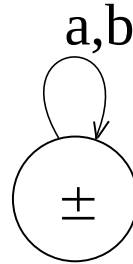
Example

- Consider the Language L , defined over $\Sigma = \{a, b\}$ of **all strings including Λ** , The language L may be accepted by the following FA



- The language L may also be accepted by the following FA

Example Continued ...

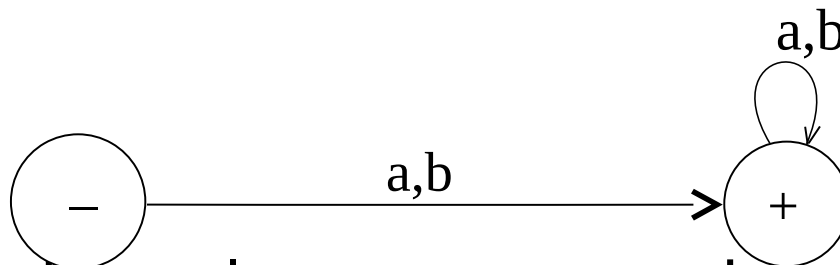


- The language L may be expressed by the following regular expression

$$(a + b)^*$$

Example

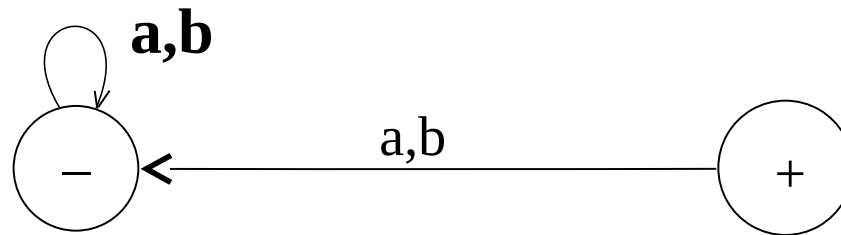
- Consider the Language L , defined over $\Sigma = \{a, b\}$ of **all non empty strings**. The language L may be accepted by the following FA



The above language may be expressed by the following regular expression $(a + b)^+$

Example

- Consider the following FA, defined over $\Sigma = \{a, b\}$



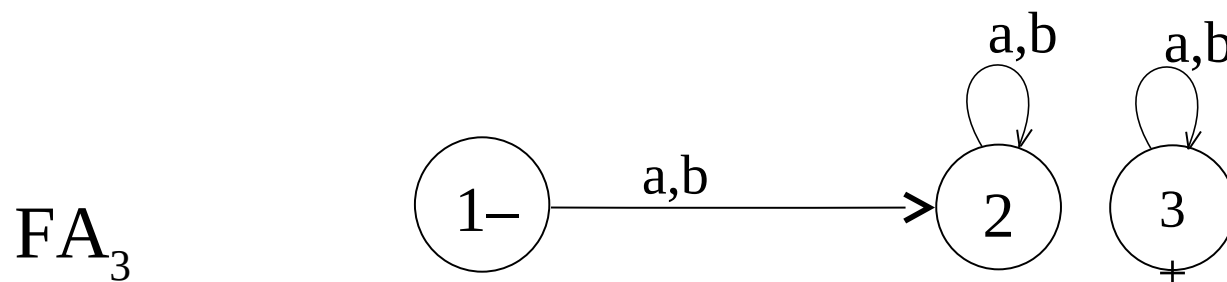
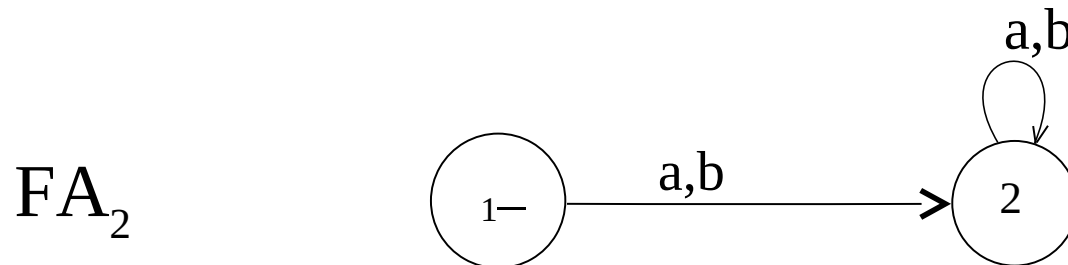
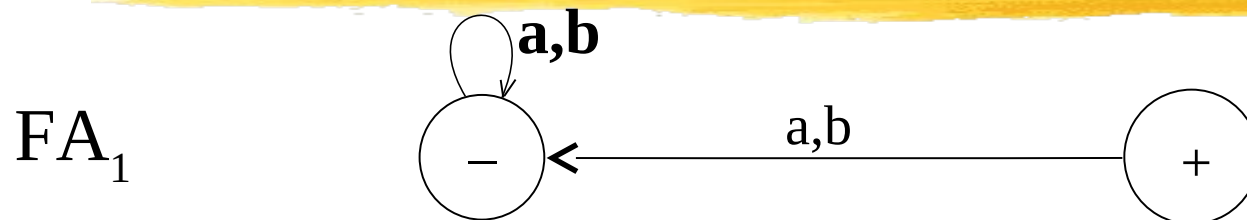
- It is to be noted that the above FA **does not accept any string**. Even it does not accept the null string. As there is no path starting from initial state and ending in final state.

Equivalent FAs



- It is to be noted that two FAs are said to be equivalent, if they accept the same language, as shown in the following FAs.

Equivalent FAs Continued ...



Note (Equivalent FAs)

- FA₁ has already been discussed, while in FA₂, there is no final state and in FA₃, there is a final state but FA₃ is disconnected as the states 2 and 3 are disconnected.

It may also be noted that the language of strings accepted by FA₁, FA₂ and FA₃ is denoted by the empty set *i.e.*

{ } OR $\emptyset$

Example

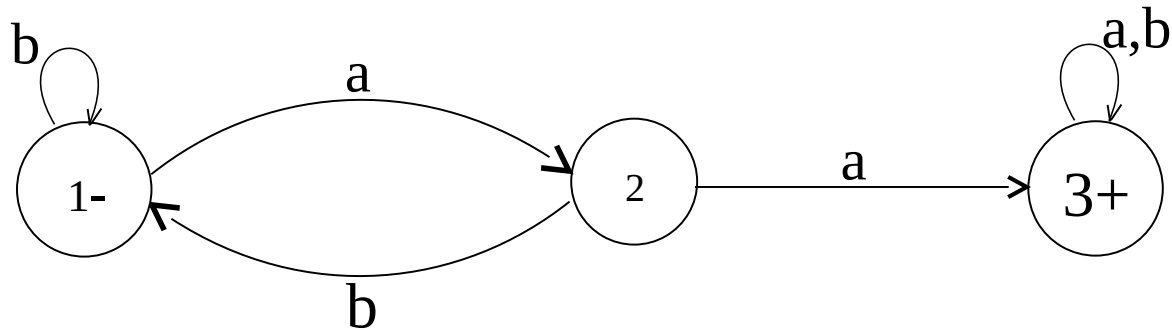
Consider the Language L of strings , defined over $\Sigma = \{a, b\}$, **containing double a.**

The language L may be expressed by the following regular expression

$$(a+b)^* (aa) (a+b)^*.$$

This language may be accepted by the following FA

Example Continued ...



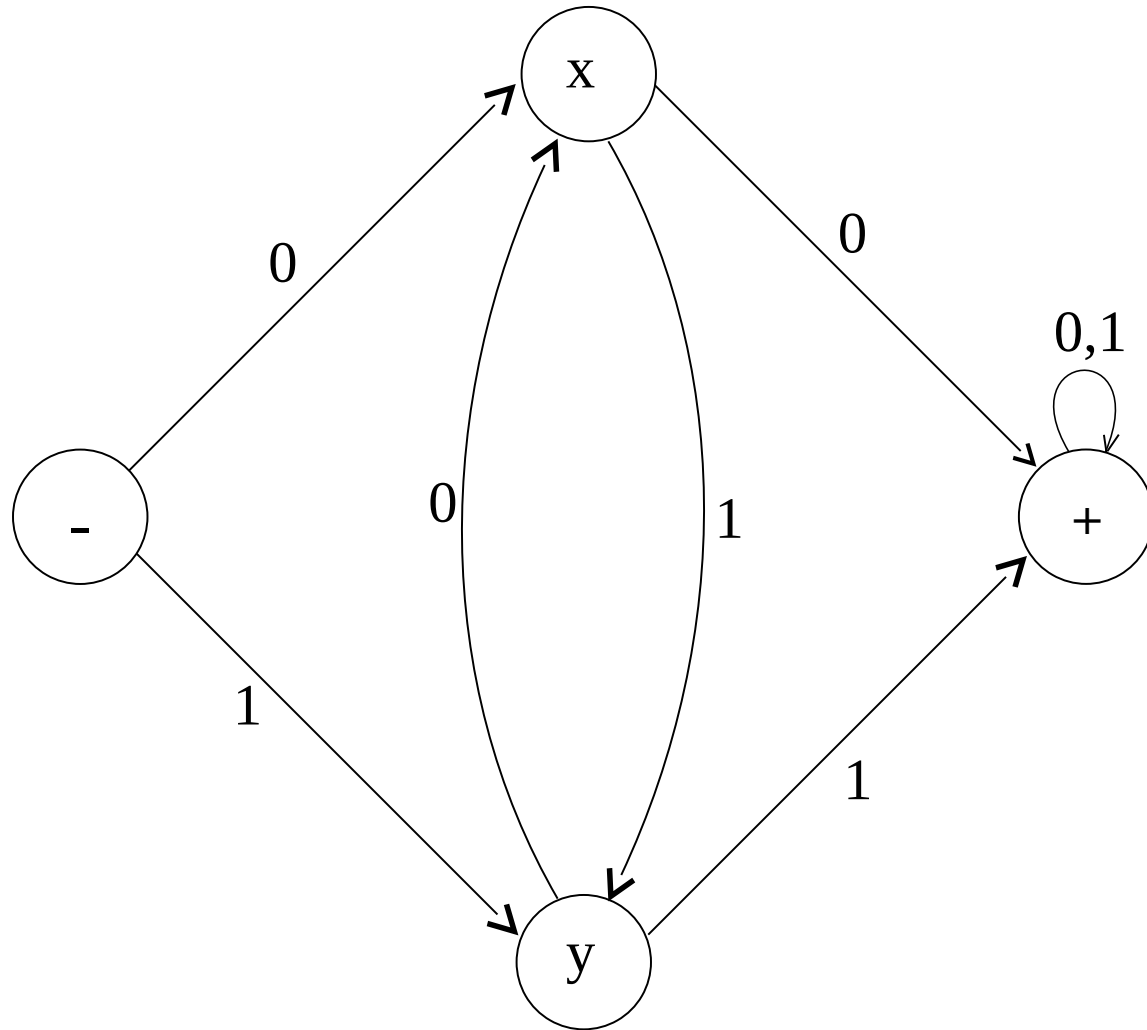
Example

Consider the language L of strings, defined over

$\Sigma = \{0, 1\}$, **having double 0's or double 1's**,
The language L may be expressed by the
regular expression $(00 + 11)(0+1)^*$

This language may be accepted by the
following FA

Example Continued ...



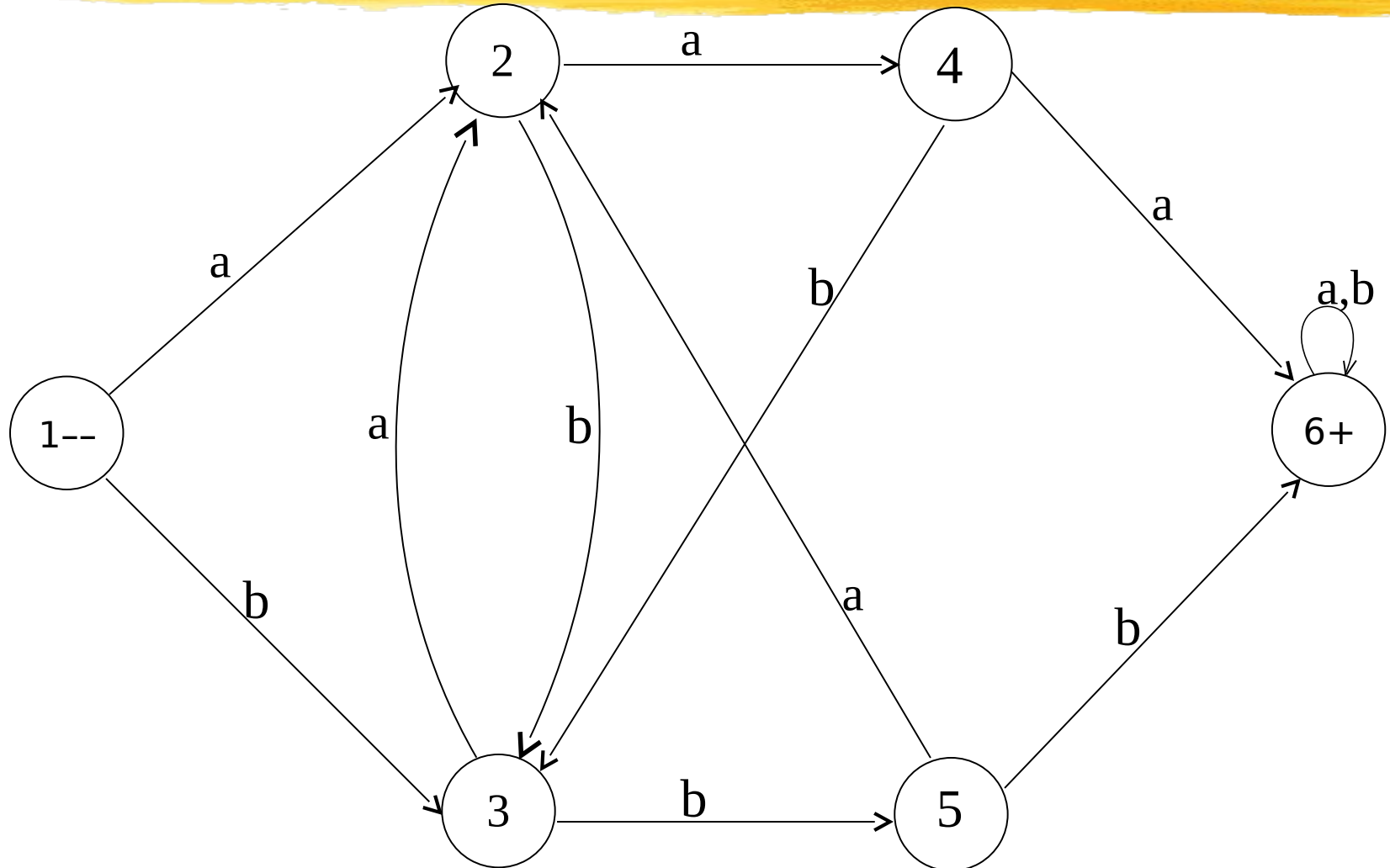
Example

Consider the language L of strings, defined over $\Sigma = \{a, b\}$, **having triple a's or triple b's**. The language L may be expressed by RE

$$(a+b)^* (aaa + bbb) (a+b)^*$$

This language may be accepted by the following FA

Example Continued ...



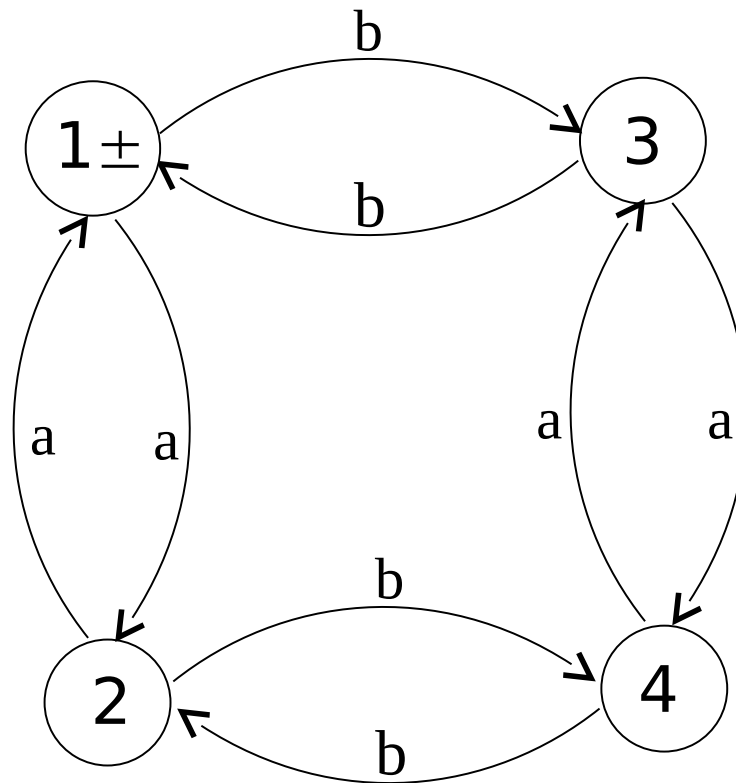
Example

Consider the **EVEN-EVEN** language, defined over $\Sigma = \{a, b\}$. As discussed earlier that **EVEN-EVEN** language can be expressed by the regular expression

$$(aa+bb+(ab+ba)(aa+bb)^*(ab+ba))^*$$

EVEN-EVEN language may be accepted by the following FA

Example Continued ...



Summing Up



Language of strings beginning with and ending in different letters, Accepting all strings, accepting non-empty strings, accepting no string, containing double a's, having double 0's or double 1's, containing triple a's or triple b's, EVEN-EVEN